Unit: Unit 0: Foundations of Algebra
Topic: Equations

Name: ____________________
Date: ____________

Homework (Carolina Reaper)

Q1. A musician is tuning her guitar. If she tightens a string to $3 \cdot x$ Hz and then loosens it to $2 \cdot x + 5$ Hz, what is x if the resulting frequency equals 11 Hz?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q2. An artist is mixing paint on a $12 \cdot x$ by $15 \cdot x$ cm canvas. If the area of the canvas is 1800 square cm, what is x?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q3. A DJ is creating a rhythm pattern with a $2 \cdot x + 1$ beat intro, followed by a $5 \cdot x - 3$ beat verse. If the total duration is 23 beats, what is x?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q4. A sound engineer is adjusting the frequency of a speaker to $x - 2$ Hz. If the new frequency is 5 Hz higher than the original 8 Hz, what is x?
(A) 10
(B) 11
(C) 12
(D) 13
(E) 14

Q5. A music producer is creating a song with a tempo of $120 + 2 \cdot x$ beats per minute. If the tempo increases by 10 beats per minute, what is x?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q6. An art gallery has a painting with a frame that is $2 \cdot x$ cm wide and $3 \cdot x$ cm tall. If the frame's area is 96 square cm, what is x?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q7. A drummer is playing a rhythm with $x + 2$ beats in the intro and $2 \cdot x - 1$ beats in the verse. If the total number of beats is 15, what is x?
(A) 3
(B) 4
(C) 5
(D) 6
(E) 7

Q8. A sound wave has a frequency of $x - 3$ Hz. If the frequency increases by 2 Hz, what is x if the new frequency is 7 Hz?
(A) 6
(B) 7
(C) 8
(D) 9
(E) 10

Open Ended Questions (FRQ)

Q9. Solve the equation $2 \cdot x + 5 = 11$ and check your solution for accuracy. Show all work.

Q10. A painter is creating a mural on a $10 \cdot x$ by $8 \cdot x$ foot wall. If the area of the wall is 480 square feet, solve for x and check your solution. Show all work.

Chef's Tips

- Tip 1: Ensure students show all work for FRQs
- Tip 2: Encourage students to check their solutions for accuracy
- Tip 3: Remind students to use proper mathematical notation